

3D Models	SmartArt	Icons	Pictures	Table Forms	Recommended	PivotTables	Tables
Checkboxes	Recommended Charts	Charts	Maps	PivotChart	Line	Column	Win/Loss
Slicer	Timeline	Link	Comments	Text	Header & Footer	Wordart	Equation
Signature	Object	Line	Symbol	Equation	Equation	Equation	Equation

This variable captures the date of the second valid tandem autologous (AUTO) SCT—where a tandem is two AUTOs > 60 and <= 180 days apart—or, if only a single AUTO occurred, the date of that single AUTO.

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LOT1_TX_AUTO_MAX_DT	Latest date of patient's valid HSCT	This variable captures the date of the second valid autologous (AUTO) SCT — where a tandem is two AUTOs >= 60 and <= 180 days apart — or, if only a single AUTO occurred, the date of that single AUTO.			Derived from T_MEDICAL.procedure code dates for autologous HSCT, after 14 day windowing and 60-day gap validation.			JM 06Apr2026
LOT1_TX_AUTO_FLAG	Flag: patient had any valid autologous HSCT	'Stem cell transplant (SCT): A LOT1-specific binary variable is created to classify whether patients have an SCT during the LOT1. This binary variable indicates whether the patient received any valid autologous transplant, regardless of single or two SCTs.			Derived flag from T_MEDICAL.procedure code analysis for autologous HSCT events.			JM 06Apr2026
LOT1_BASE_MEDS	LOT1 induction medications	The LOT1 regimen is defined as all multiple myeloma (MM) oncology therapies received on and within 60 days of the LOT1 start date, with the first therapy received considered the induction therapy. This variable therefore captures the full set of oncology agents administered during this 60-day window, excluding non-oncology agents such as corticosteroids. Switches between a biologic reference product and its biosimilars(s), or between biosimilars of the same reference product, are permitted and do not constitute a change in regimen or advance the line of therapy (Madduri, 2021).			Medications identified from T_MEDICAL (PROC_CD, BILL_PROC_CD, NDC) and T_RX (NDC) matched against CL_MMA_CODELIST (Tab 41). Medication abbreviations from CL_MMA_ROLLUP (Tab 40). Claims filtered by MAP_START_DT within induction window from LOT1_START_DT.			JM 06Apr2026
LOT1_BASEMAINT_START	Start date of maintenance period	A maintenance regimen is a period of 120 days or longer during which only a valid maintenance therapy is available. Maintenance is not required to be prescribed continuously for the duration of the 120 day period. Furthermore, the maintenance therapy must have been a part of the LOT1 initial regimen.			Maintenance medications identified from T_MEDICAL.T_RX claims matched to CL_MMA_ROLLUP medications with MONOMAINTEANCE=1 or valid DUALMAINTENANCEWITH combinations. Start date is the first MAP_START_DT of the maintenance medication after LOT1_BASE_DISCON_DT (pre-maintenance discontinuation).			
LOT1_BASEMAINT_TYP	Maintenance medication abbreviation	valid maintenance therapies per NCCN Guidelines include lenalidomide monotherapy, bortezomib monotherapy, daratumumab monotherapy, ixazomib monotherapy, thalidomide monotherapy, bortezomib/lenalidomide dual, carfilzomib/lenalidomide dual, and daratumumab/lenalidomide dual.			Medication abbreviation from CL_MMA_ROLLUP (Tab 40) MED_ABBR field, identifying the maintenance drug from T_RX (NDC) or T_MEDICAL (PROC_CD/NDC) claims.			
LOT1_BASEMAINT_END	End date of maintenance period (temp)	Once a maintenance regimen has begun, it continues until discontinuation, presence of an SCT, right censoring (i.e., end of follow-up), or the addition of any non-maintenance MM therapy automatically causes the maintenance regimen to end.			Maintenance end date derived from MAP_END_DT for maintenance medications in T_MEDICAL.T_RX, compared against other end events from the same source tables.			
LOT1_BASEMAINT_MED_LIST	Flag for maintenance medication [MED] (BORTIDARAIXAZLE NACARR/THAL)	valid maintenance therapies per NCCN Guidelines (Kumar, 2020) include mono-maintenance (lenalidomide, bortezomib, daratumumab, ixazomib, thalidomide) and dual-maintenance (bortezomib/lenalidomide, carfilzomib/lenalidomide, daratumumab/lenalidomide). 'Any of the maintenance medications listed above will be considered maintenance medications for all patients, regardless of transplant status.'			Individual medication flags derived from T_RX (NDC) and T_MEDICAL (PROC_CD/NDC) claims matched to CL_MMA_CODELIST (Tab 41) and CL_MMA_ROLLUP (Tab 40) for specific medication abbreviations.			
LOT1_BASEMAINT_END_REASON	Maintenance period end reason	The LOT1 will end after a maintenance regimen ends due to: unplanned SCTs, death, health plan disenrollment, and end of study period. The addition of any non-maintenance MM therapy automatically causes the maintenance regimen to end.			End reason derived from the same Optum CDM source tables as LOT1_END_REASON_TEMP: T_MEDICAL.T_RX for medication events, T_DOD for death, T_MEMBER.CONT_ENROLLMENT for disenrollment.			
LOT1_BASE_END_DT	Final LOT1 base period end date	The LOT1 will continue until the earliest of any of the following (Rules 2-8 are ending events; Rule 1 from the protocol defines permissible substitutions that do NOT end the LOT1). Per Rule 2: 'The LOT end date is based on the run-out date, defined as the last date that any MM medication in that LOT is considered to be available.' If the current LOT is immediately interrupted by a new agent, the LOT end date is the day before the first administration/discontinue date of the new agent.			LOT1_BASE_END_DT derived from priority evaluation of dates from T_MEDICAL.T_RX (MAP_END_DT for discontinuation, MAP_START_DT for new med), T_MEDICAL.procedure codes (SCT events), T_DOD (YMOD for death), T_MEMBER.CONT_ENROLLMENT (EUEEND for disenrollment), and study_end config parameter.			JM 06Apr2026
LOT1_BASE_END_REASON	Final LOT1 base period end reason	LOT ends for one of the following reasons (Rules 2-8 from the protocol): (2) Discontinuation of all agents, (3) Unplanned SCTs; (4) SCTs not followed by maintenance within 180 days; (5) Death; (6) Health plan disenrollment; (7) End of study period; (8) End of maintenance regimen, Rule 1 (permissible substitutions - biologic reference product with biosimilars) does not advance the LOT and is not an ending event.			End reason derived from evaluation of dates across T_MEDICAL.T_RX (medication events), T_MEDICAL (SCT procedure codes), T_DOD (death), and T_MEMBER.CONT_ENROLLMENT (disenrollment). Priority logic applied analytically.			JM 06Apr2026

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 IOT ends for one of the following reasons (Rules 2-8 from the protocol): (2) Discontinuation of all agents; (3) Unplanned SCIs; (4) SCIs not followed by maintenance within 180 days; (5) Death; (6) Health plan disenrollment; (7) End of study period; (8) End of maintenance regimen. Rule 1 (permissible substitutions - biologic reference product with biosimilars) does not advance the IOT and is not an ending event.

1	LOT1_BASE_END_REASON	Final LOT1 base period end reason		LOT ends for one of the following reasons (Rules 2-8 from the protocol): (2) Discontinuation of all agents; (3) Unplanned SCTs; (4) SCTs not followed by maintenance within 180 days; (5) Death; (6) Health plan disenrollment; (7) End of study period; (8) End of maintenance regimen. Rule 1 (permissible substitutions - biologic reference product with biosimilars) does not advance the LOT and is not an ending event.		End reason derived from evaluation of dates across T_MEDICAL_TX (medication events), T_MEDICAL (SCT procedure codes), T_DOD (death), and T_MEMBER_CONT_ENROLLMENT (disenrollment). Priority logic applied analytically.	JM 06Apr2026
2	FIRST_ALLO_DT	Date of first allogeneic HSCT during LOT1		Allogeneic SCT: Presence of an allogeneic SCT immediately ends a LOT and starts a new LOT. Rule 3: Any allogeneic SCTs are considered a new LOT and the current LOT will end the day before an allogeneic SCT occurs. *Allogeneic SCT: Presence of an allogeneic SCT is classified as its own LOT, with no other MM therapies included in that LOT.		ALLO SCT identified from T_MEDICAL procedure codes matching CL_SCT_CODELIST entries where SCT_TYPE=ALLO. Date from FST_DT in T_MEDICAL. Sequential ordering applied to identify first ALLO event.	JM 06Apr2026
3	FIRST_CAR_T_DT	Date of first CAR-T therapy during LOT1		CAR-T: CAR-T cellular therapy infusions are classified as their own LOT. Therapies (including supportive agents, e.g., corticosteroids) given within 45 days of CAR-T cellular therapy are consolidated as part of the CAR-T LOT.		CAR-T identified from T_MEDICAL procedure codes matching CL_SCT_CODELIST entries where SCT_TYPE=CART. Date from FST_DT in T_MEDICAL.	
4	LOT1_DISCON_GAP_DAYS	Discontinuation gap threshold (days)		Discontinuation of the LOT1 regimen occurs when all MM oncology agents in the line of therapy are stopped. If the regimen is discontinued and no new agent is initiated, the LOT end date is defined as the run-out date of the last administered agent. Sensitivity analyses will assess the robustness of the line of therapy rules by varying key algorithm parameters		Gap calculated as datediff(OBS_END_DT, RAW_DISCON_DT) where OBS_END_DT is derived from T_MEMBER_CONT_ENROLLMENT (ELIGEND), T_DOD (NMDD), and study_end config. RAW_DISCON_DT is max(MAP_END_DT) from T_MEDICAL_TX for induction medications.	
5	MEDICAL_DAY_SUPPLY	Assumed days supply for medical claims		The days supply on medical claims is always missing and therefore a 28-day supply is assumed. When pharmacy claims are missing days supply, or when days supply values are anomalous, a 28-day supply is assumed.		Applied to T_MEDICAL claims where DAYS_SUPPLY is NULL. For TX pharmacy claims, DAYS_SUPPLY is typically populated but when missing/anomalous, 28 days is substituted. The assumed supply extends MAP_END_DT - FST_DT (or FILL_DT) + days_supply - 1.	JM 06Apr2026
6	LOT1_BASE_LENGTH	LOT1 base period duration in days		The protocol does not define LOT1_BASE_LENGTH as a named variable. Duration is implied by Protocol Section 5.1.1 which defines LOT start date (LOT1 begins on the date of the first fill for an MM therapy following a patient's index date) and LOT end date (earliest of Rules 1-8 in the protocol).		Duration computed from dates already derived from T_MEDICAL_TX (LOT1_START_DT from first MAP_START_DT) and LOT1_BASE_END_DT (priority-based end date).	JM 06Apr2026
7	LOT1_SCT_TO_MAINT_180D_RULE	SCT not followed by maintenance within 180 days - LOT end rule		SCTs not followed by maintenance: If an SCT is not followed by a maintenance regimen within 180 days, then the last day of the LOT is the date of the SCT. *Following an autologous SCT (within 180 days of the SCT or in the case of a tandem autologous SCT within 180 days of the second SCT). The expectation after an SCT is for a maintenance period and thus instead of the usual 120 days, we check for a valid 30-day (minimum) maintenance period.		Would require checking T_MEDICAL_TX claims for maintenance medications (NONMAINTENANCE=1 or valid DUALMAINTENANCEWITH in CL_MMA_ROLLUP) within 180 days after the SCT date (LOT1_1ST_SCT_DT or LOT1_TX_AUTO_MAX_DT).	
8	PERMISSIBLE_SUBS_EFFECT	Permissible substitution rules affecting LOT1 end		*Substitution of a biologic reference product with any of its biosimilars, or between biosimilars of the same reference product, does not advance the LOT; such permissible substitutions are considered within the same line, which is named according to the drug given for the longest duration.		Substitutions tracked via CL_MMA_ROLLUP (Tab 40) which maps medication abbreviations. The permissible_sub reference table pairs original_med with substitute_med. When a new medication claim appears in T_MEDICAL_TX, it is checked against permissible_subs before being classified as a new medication addition.	JM 06Apr2026
9	LOT1_TX_ENDDATE_REASON	SCT end reason type (AUTO/ALLO/CART)		The type of stem cell transplant (SCT) determines how LOT1 ends. In the case of an autologous SCT, an unplanned transplant ends the current line of therapy the day before the SCT date. For an allogeneic SCT, the current LOT ends the day before the allogeneic SCT date, and a new LOT begins from the date of the transplant. CAR-T cellular therapy infusions are classified as their own line of therapy, with the preceding LOT ending the day before the CAR-T infusion date; however, if a new oncology agent is introduced within 45 days of the CAR-T infusion, that agent is incorporated into the CAR-T LOT rather than initiating a new line of		SCT type determined from T_MEDICAL procedure codes matched against CL_SCT_CODELIST (Tab 44) which classifies procedures as AUTO, ALLO, or CART. Priority is earliest LOT-ending date wins.	

